# Data Handling

4DT910/4DV510: Data Mining

# Outline

- Attributes and Objects
- Types of Datasets
- Quality of the Data
- Similarity and Distance
- Data Exploration
- Data Preprocessing



*Based on lecture materials and Chapter 2 (“Data”) in Introduction to Data Mining, 2nd Edition, by Tan, Steinbach, Karpatne, and Kumar.*

# What is Data?

- Collections of ***data objects*** and their ***attributes***
  - Other names: *record, point, observation, vector, sample, etc*.
- ***An attribute*** is a property or characteristic of an object
  - Other names: *variable, field, feature, or dimension*.

Attributes

Objects

| ID | Age (years) | Gender | Occupation | Annual Income (KSEK) | City |
|---|---|---|---|---|---|
| 1 | 23 | F | Student | 190 | Växjö |
| 2 | 34 | M | Engineer | 420 | Kalmar |
| 3 | 45 | F | Nurse | 360 | Lund |
| 4 | 28 | M | Teacher | 310 | Växjö |
| 5 | 52 | F | Manager | 550 | Kalmar |
| 6 | 31 | M | Designer | 340 | Lund |
| 7 | 41 | F | Accountant | 390 | Växjö |
| 8 | 37 | M | Technician | 290 | Kalmar |

# Attribute Values

- **Attribute values** are numbers or symbols assigned to an attribute for a particular object
  - *Example*: For a student, Age = 23 and City = Växjö.
- Distinction between attributes and attribute values
  - Same attribute can be represented using different values or units
    - *Example*: height can be measured in feet or meters
  - Different attributes can use the same type of values but have different meanings
    - *Example*: Attribute values for Customer ID=25 and age=25 are integers

# Types of Attributes

| | Attribute Type | Description | Examples | Operations |
|---|---|---|---|---|
| Categorical Qualitative | Nominal | The values of a nominal attribute are just different names. | zip codes, employee ID numbers, eye color, sex: {*male*, *female*} | mode, entropy, contingency correlation, $\chi^2$ test |
| | Ordinal | The values of an ordinal attribute provide enough information to order objects. | hardness of minerals, {*good, better, best*}, grades, street numbers | median, percentiles, rank correlation, run tests, sign tests |
| Numeric Quantitative | Interval | For interval attributes, the difference between values are meaningful. | calendar dates, temperature in Celsius or Fahrenheit | mean, standard deviation, Pearson's correlation, *t* and *F* tests |
| | Ratio | For ratio variables both difference and ratios are meaningful. | temperature in Kelvin, monetary quantities, counts, age, mass, length, current | geometric mean, harmonic mean, percent variation |

The attribute type determines which comparisons and operations are meaningful:

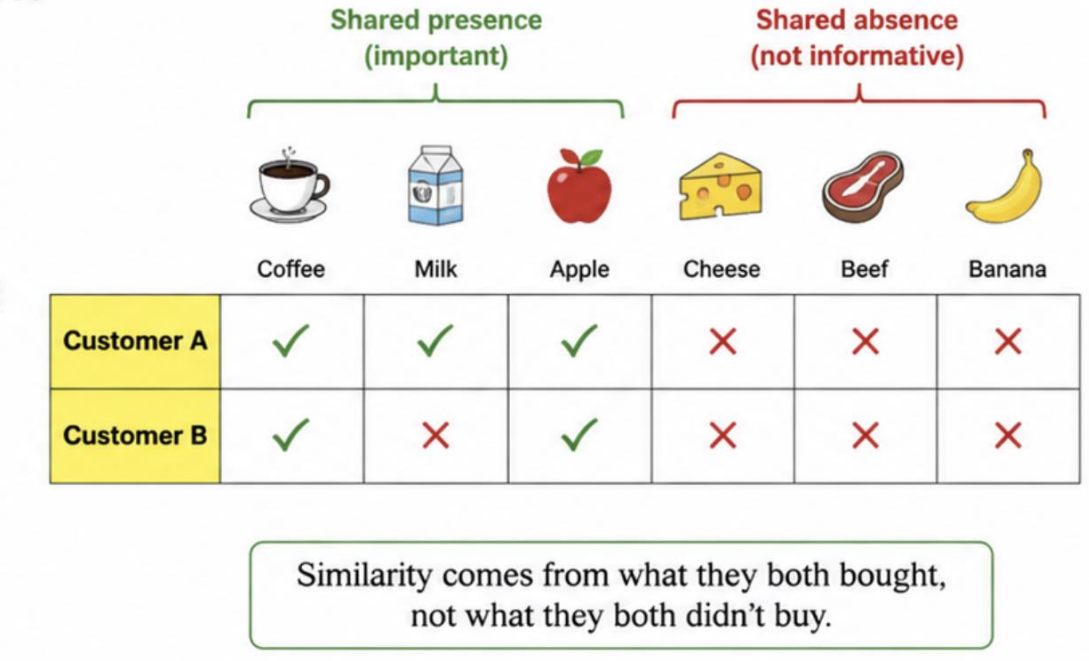
- **Distinctness:** =, ≠
- **Order:** <, >
- **Differences:** +, −
- **Ratios:** ×, ÷

# Describing Attributes be the Number of Values

- **Discrete Attribute**
  - Has only a finite or countably infinite set of values
  - *Examples*: zip codes, counts, or the set of words in a collection of documents
  - Often represented as integer variables.
  - Note: binary attributes are a special case of discrete attributes
- **Continuous Attribute**
  - Has real numbers as attribute values
  - *Examples*: temperature, height, or weight.
  - Practically, real values can only be measured and represented using a finite number of digits.
  - Continuous attributes are typically represented as floating point variables.

# Asymmetric Attributes

- An **asymmetric attribute** is a binary attribute where **presence (1) is more important than absence (0)**.
  - *Example:* Two customers buying the same product indicates similarity; both not buying a product usually does not.
  - Used in: transactions, text data, medical data, and recommendation systems.



# Limitations of Attribute Categorization

- Incomplete
  - asymmetric binary, cyclical, multivariate, partially ordered data, etc.
- Real data can be approximate
  - This can complicate recognition of the proper attribute type
    - *Example:* Age can be recorded exactly (35 years → ratio) or in age groups (30–39 → ordinal), making the appropriate attribute type dependent on its representation.
  - Treating one attribute type as another may be approximately correct
    - *Example:* A 1–5 rating is ordinal, but it is often treated as numeric to calculate an average rating.
- Real data can be noisy
  - Measurements may contain errors or random variation
    - *Example:* A sensor may record slightly different values for the same underlying condition

# Types of Data

# Characteristics of Data

- **Dimensionality** – number of attributes/features
  - High-dimensional data can be more difficult to analyze and visualize.
- **Sparsity** – most attribute values are zero or absent
  - Common when objects contain only a small subset of all possible features.
  - *Example:* A patient takes 5 medications out of 1,000 possible medications → most medication attributes are 0 (not taken).
- **Resolution** – level of detail at which data are measured
  - Different patterns may appear at different levels of resolution.
- **Size** – number of data objects/records
  - Dataset size can determine which algorithms and computational methods are practical.
  - *Example:* Small datasets may favor simpler models and carefully selected features over deep learning.

# Common Types of Datasets

- Tabular/Record
  - Data Matrix
  - Transaction Data
  - Document Data
- Graph/Network
  - Social netwroks, web pages links, biological networks
- Ordered
  - Spatial Data (e.g., geolocation)
  - Temporal Data (e.g., sensors measurements over time)
  - Sequential Data (e.g., event-based data)

# Tabular/Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

| ID | Age | Gender | Name |
|---|---|---|---|
| P001 | 34 | Female | Anna |
| P002 | 58 | Male | Erik |
| P003 | 27 | Female | Sara |
| P004 | 63 | Male | Johan |
| P005 | 45 | Female | Karin |

# Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such a data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

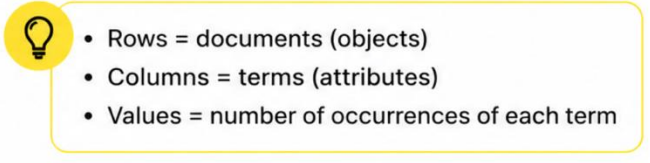
**Data Matrix**

| Age | Height | Weight | Heart Rate |
|---|---|---|---|
| 34 | 168 | 65 | 72 |
| 58 | 175 | 82 | 78 |

# Document Data

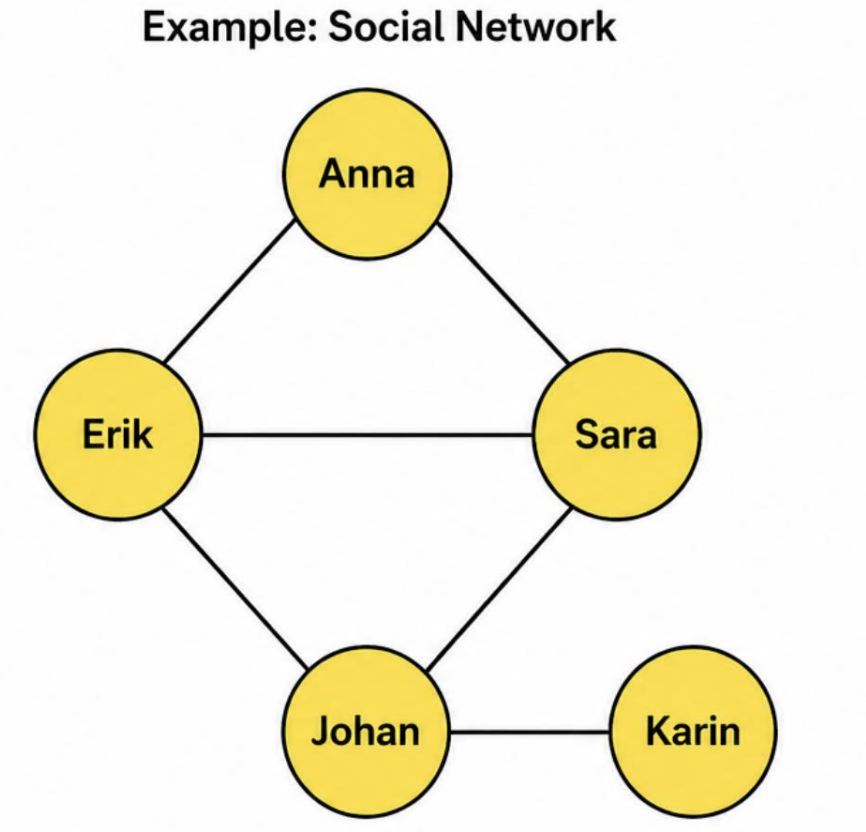
- Documents contain unstructured text.
- Can be stored as .txt files, text fields in CSV files, databases, etc.
- To apply many data-mining methods, the text can be represented as a document–term matrix (part of the preprocessing step).

| Document | patient | reports | pain | nausea |
|---|---|---|---|---|
| D1 | 1 | 1 | 1 | 0 |
| D2 | 1 | 1 | 0 | 1 |
| D3 | 1 | 0 | 1 | 1 |



# Graph Data

- Graph data represent objects and the relationship between them
- A graph consist of:
  - Nodes (vertices) = objects
  - Edges (links) = relationships between objects
- Nodes and edges can also have attributes



# Ordered Data

- Ordered data have a natural order or time/space dependency
- Observations are collected over time, and the order carries important information

**Example: Sensor Measurements (Wearable Device)**

A wearable device records accelerometer and gyroscope data over time at 100 Hz.

| Timestamp (s) | Acc X ($m/s^2$) | Acc Y ($m/s^2$) | Acc Z ($m/s^2$) | Gyro X (°/s) | Gyro Y (°/s) | Gyro Z (°/s) |
|---|---|---|---|---|---|---|
| 00.00 | 0.12 | -0.03 | 9.81 | 0.5 | -0.2 | 0.1 |
| 00.01 | 0.15 | -0.01 | 9.79 | 0.6 | -0.1 | 0.2 |
| 00.02 | 0.18 | 0.00 | 9.78 | 0.7 | 0.0 | 0.3 |
| 00.03 | 0.20 | 0.02 | 9.76 | 0.8 | 0.1 | 0.4 |
| 00.04 | 0.17 | 0.01 | 9.77 | 0.8 | 0.1 | 0.4 |
| ... | ... | ... | ... | ... | ... | ... |
| 10.00 | -0.05 | 0.03 | 9.74 | -0.2 | 0.3 | -0.1 |

# Data Quality

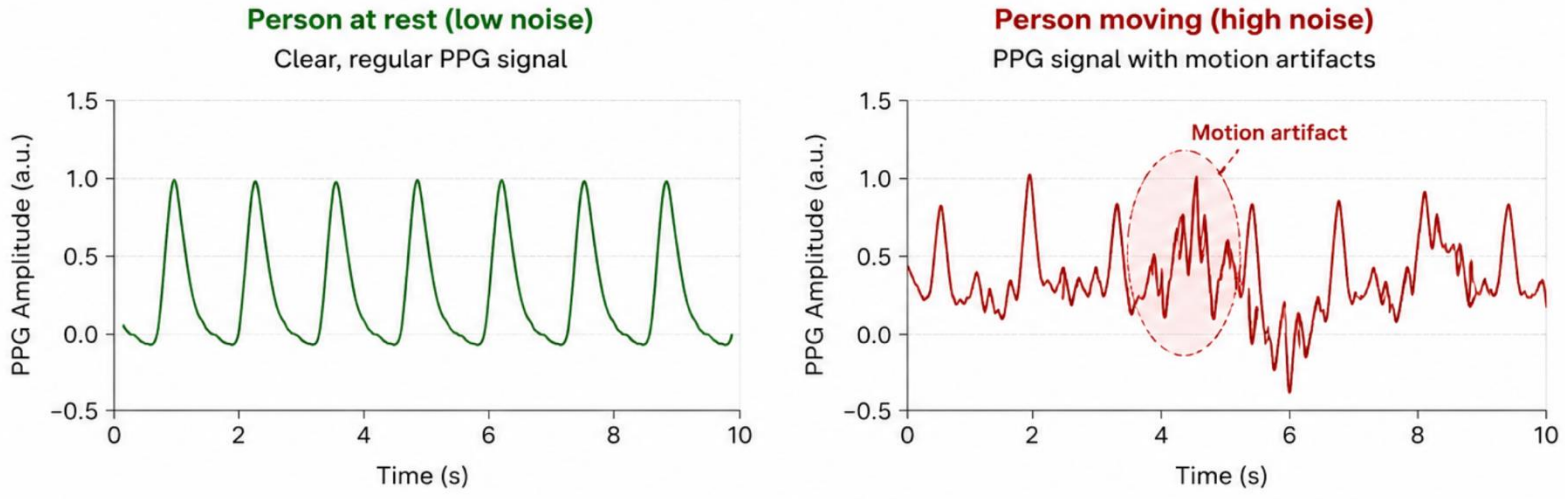
# Data Quality

- **Data quality** describes how accurate, complete, consistent, and reliable data are for the intended analysis.
- Real-world datasets often contain **data quality problems**, such as noise, missing values, incorrect values, outliers, duplicate data,e tc.
- Before analysis, we need to consider:
  - How can we detect these problems?
  - What should we do about them?
- **Two approaches:** Improve the data through **data cleaning**, or use methods that are robust to imperfect data.

# Noise

- Noise is random variation that distorts the data
- In sensor data, noise can come from the environment, the device, or the person (e.g., motion).
  - Example: PPG (Photoplethysmography) signal

PPG measures blood volume changes. The clean signal is visible when the person is still. Motion introduces artifacts (noise) that distort the signal.



# Outliers

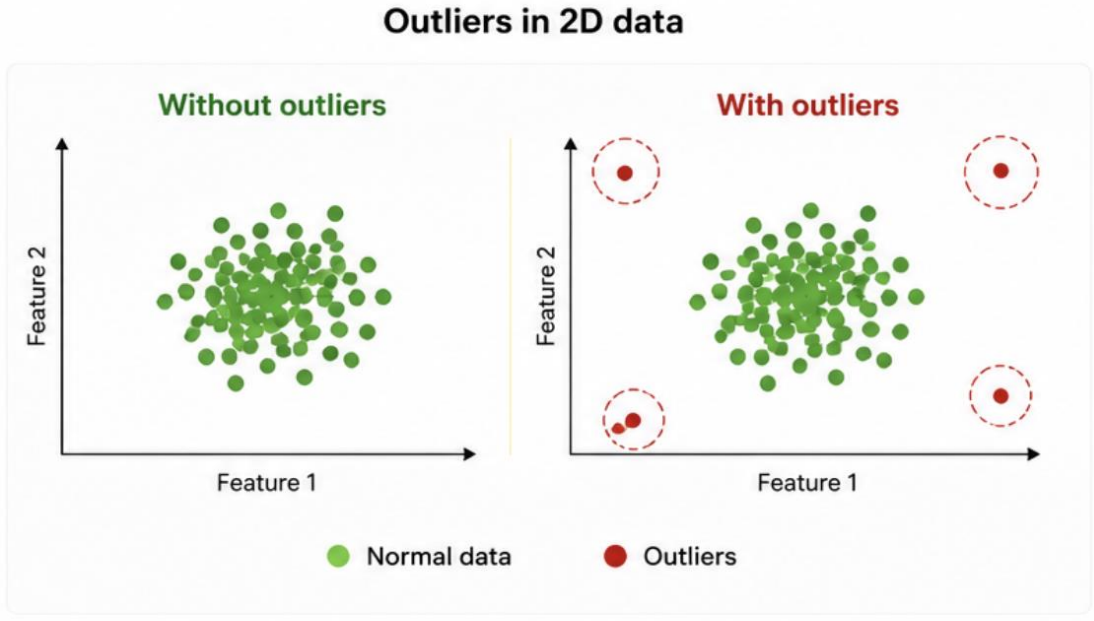
Outliers are data objects with characteristics that are **considerably different** than most of the other data objects in the data set.

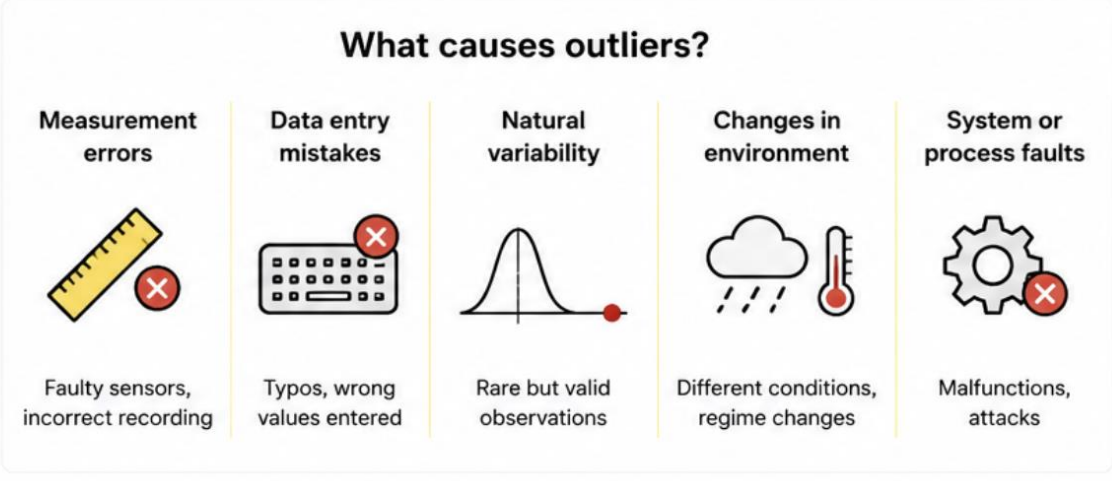
## 1 Case 1: Outliers are noise that interferes with data analysis

- Can distort statistics and models
- Ideally detected and handled (e.g., removed, corrected, or robust methods used)

## 2 Case 2: Outliers are the goal of our analysis

- Represent rare but important events
- Examples:
  - Credit card fraud
  - Intrusion detection



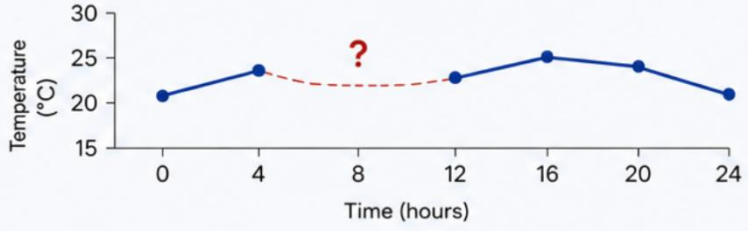
# Missing Values

Missing values occur when a value for an attribute is **not available** for a data object.

## Why are values missing?

- **Information was not collected**
  (e.g., a patient did not report weight)
- **Measurement or recording failed**
  (e.g., a sensor temporarily stopped recording)
- **An attribute is not applicable**
  (e.g., pregnancy status for some patients)
- **Information was lost or incorrectly entered**
  (e.g., data entry errors)

### Example in time series (temperature)




--? = Missing measurement (e.g., sensor failure)

### Example: Patient Data

| Patient | Age (years) | Weight (kg) | Heart Rate (bpm) |
|---|---|---|---|
| P1 | 35 | 72 | 68 |
| P2 | 42 | ? | 74 |
| P3 | ? | 81 | 71 |
| P4 | 56 | 76 | ? |
| P5 | 29 | 65 | 69 |

? = Missing value

## Handling Missing Values

- **Remove** objects or attributes with too many missing values
  (may reduce information)
- **Estimate / impute missing values**
  - Example: mean/median, regression, k-NN
  - Example: interpolation for time-series data
- **Ignore** missing values when the analysis method allows it
  (e.g., pairwise deletion in correlations)
- **Use algorithms that can handle missing values directly**
  (e.g., tree-based models like XGBoost, LightGBM)

**Important:** There is no single best solution — how we handle missing values depends on why the values are missing, how many are missing, and the analysis we want to perform.

# Duplicated Data

- **Duplicate data** occur when the same object or information is recorded more than once.
- Common causes:
  - Merging data from different sources
  - Repeated records or data imports

**Example: Medication data**

| Patient ID | Medication | ATC code | Source |
|---|---|---|---|
| P01 | Metformin | A10BA02 | Dataset A |
| P01 | Metformin | A10BA02 | Dataset B |
| P01 | Amlodipine | C08CA01 | Dataset A |
| P01 | Amlodipine | C08CA01 | Dataset B |

**4 records** → but only **2 unique** medications

To count simultaneous medications, use **unique ATC codes**.

## Should duplicates always be removed?

**No.** Repeated records may represent real events.

**Example:** The same medication prescribed on two different dates.

| Patient ID | Medication | ATC code | Date | Meaning |
|---|---|---|---|---|
| P01 | Metformin | A10BA02 | Jan 10 | Prescription issued |
| P01 | Metformin | A10BA02 | Mar 15 | Prescription refill |

# Similarity and Dissimilarity Measures

# Similarity and Dissimilarity Measures

- **Similarity** measure
  - Numerical measure of how alike two data objects are.
  - Is higher when objects are more alike.
  - Often falls in the range [0,1]
- **Dissimilarity** measure
  - Numerical measure of how different two data objects are
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
  - Upper limit varies
- **Proximity** refers to a similarity or dissimilarity

# Similarity/Dissimilarity for Simple Attributes

- The following table shows the similarity and dissimilarity between two objects, x and y, with respect to a single, simple attribute.

| Attribute Type | Dissimilarity (distance *d*) | Similarity (s) |
|---|---|---|
| **Nominal**<br>Categories, no order | $d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$ | $s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$ |
| **Ordinal**<br>Ordered categories | $d = \frac{\lvert x - y \rvert}{n - 1}$<br>(values mapped to integers 0 to $n-1$, where $n$ is the number of values) | $s = 1 - d$ |
| **Interval or Ratio**<br>Numeric values, meaningful differences | $d = \lvert x - y \rvert$ | $s = -d, \quad s = \frac{1}{1+d}, \quad s = e^{-d},$<br>or<br>$s = 1 - \frac{d - \min}{\max - \min}$ |

**Note:** The choice of similarity/dissimilarity measure depends on the type of attribute and the analysis method.

# Distance Between Data Objects

| Distance | Formula | Normalisation requires (sensitive for outliers) |
|---|---|---|
| **Euclidean Distance** | $d(x,y) = \sqrt{\sum_{k=1}^{p} (x_k - y_k)^2}$ | Yes (sensitive) |
| **Manhattan (City Block) Distance** | $d(x,y) = \sum_{k=1}^{p} \lvert x_k - y_k \rvert$ | Yes (less sensitive) |
| **Minkowski Distance (p)** | $d(x,y) = \left( \sum_{k=1}^{p} \lvert x_k - y_k \rvert^p \right)^{1/p}, p \geq 1$ | Yes (sensitive for small $p$) |
| **Chebyshev Distance** | $d(x,y) = \max_k \lvert x_k - y_k \rvert$ | No (robust) |
| **Cosine Distance** | $d(x,y) = 1 - \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\lVert \boldsymbol{x} \rVert \ \lVert \boldsymbol{y} \rVert}$ | No (robust) |

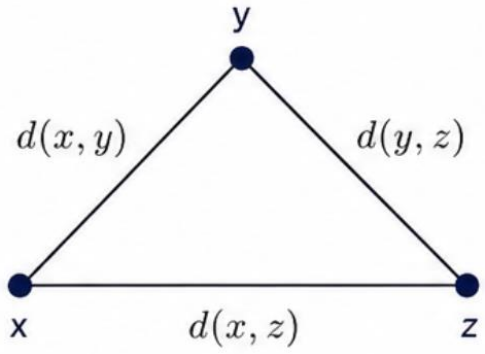
**Note:** Normalisation is recommended when features are on different scales. It is especially important for distances sensitive to outliers (e.g., Euclidean).

# Common Properties of a Distance

- A distance function d(x,y) is a metric if it satisfies the following properties

| Property | Meaning | Formula |
|---|---|---|
| **1. Non-negativity** | Distance cannot be negative. | $d(x, y) \geq 0$ |
| **2. Identity** | Distance is zero only for the same object. | $d(x, y) = 0 \iff x = y$ |
| **3. Symmetry** | Distance is the same in both directions. | $d(x, y) = d(y, x)$ |
| **4. Triangle inequality** | The direct distance cannot be longer than going via another point. | $d(x, z) \leq d(x, y) + d(y, z)$ |

**Triangle inequality (visual)**




Direct path $d(x, z)$ is not longer than path via $y$:

$$d(x, z) \leq d(x, y) + d(y, z)$$

# Similarity Between Data Objects

- Similarity measures how alike two data objects (records) are. Values are typically in the range [0,1], where 1 means identical.

| Similarity Measure | Formula | Normalisation requires (sensitive for outliers) |
|---|---|---|
| Cosine Similarity | $s(x,y) = \frac{x \cdot y}{\|\|x\|\| \ \|\|y\|\|}$ | No (robust) |
| Pearson Correlation Similarity | $s(x,y) = \frac{\sum_k (x_k - \bar{x})(y_k - \bar{y})}{\sqrt{\sum_k (x_k - \bar{x})^2}\sqrt{\sum_k (y_k - \bar{y})^2}}$ | Yes (sensitive) |
| Jaccard Similarity (for sets) | $s(x,y) = \frac{\|x \cap y\|}{\|x \cup y\|}$ | No (robust) |
| Overlap Coefficient (for sets) | $s(x,y) = \frac{\|x \cap y\|}{\min(\|x\|, \|y\|)}$ | No (robust) |
| Normalized Euclidean Similarity | $s(x,y) = \frac{1}{1 + d_{\text{norm}}(x,y)}$ | Yes (sensitive) |

**Note:** If you already have a distance $d(x,y)$, a common way to convert it to similarity is $s(x,y) = \frac{1}{1+d(x,y)}$ (range $(0,1]$, higher means more similar).

## Properties of Similarities

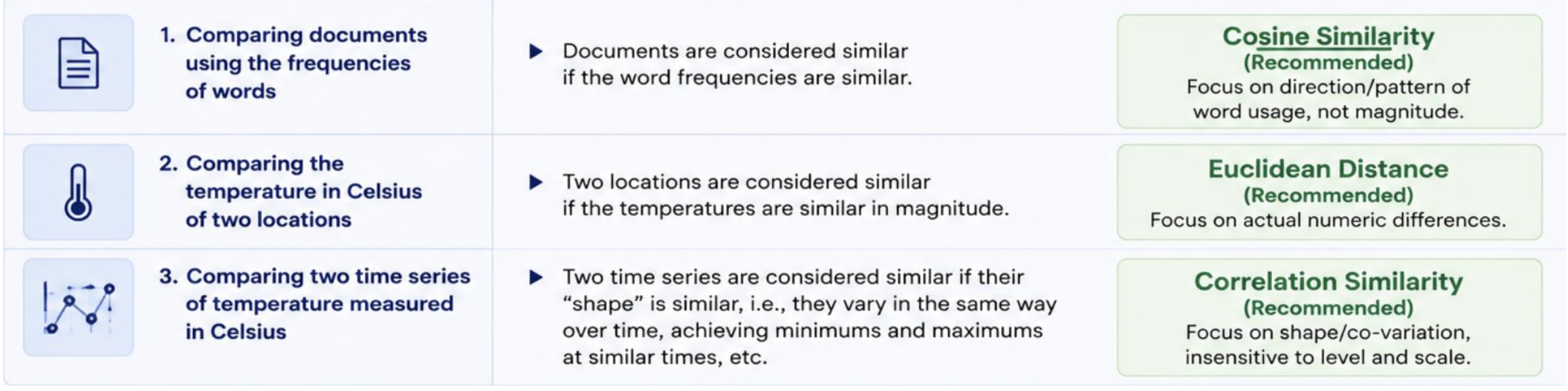
A similarity function $s(x,y)$ typically satisfies the following properties:

1. **Maximum similarity for identical objects**
   $s(x,y) = 1$ (or maximum similarity) only if $x = y$.
   (does not always hold, e.g., cosine)

2. **Symmetry**
   $s(x,y) = s(y,x)$ for all $x$ and $y$.
   (Symmetry)

# Common Properties of a Similarity

- Choice of the right proximity measure depends on the domain




Choose the proximity measure that matches what "similar" means in your domain: orientation (cosine), co-variation/shape (correlation) or absolute magnitude (Euclidean).
The measure must be applicable to the data and produce results that agree with domain knowledge

# General Approach for Combining Similarities

- Sometimes attributes are of many different types, but an overall similarity is needed.

1: For the $k^{\text{th}}$ attribute, compute a similarity, $s_k(\mathbf{x}, \mathbf{y})$, in the range [0, 1].

2: Define an indicator variable, $\delta_k$, for the $k^{\text{th}}$ attribute as follows:

$\delta_k$ = 0 if the $k^{\text{th}}$ attribute is an asymmetric attribute and both objects have a value of 0, or if one of the objects has a missing value for the kth attribute

$\delta_k$ = 1 otherwise

3. Compute $$\text{similarity}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^{n} \delta_k s_k(\mathbf{x}, \mathbf{y})}{\sum_{k=1}^{n} \delta_k}$$

# Using Weights to Combine Similarities

- May not want to treat all attributes the same.
  - Use non-negative weights $\omega_k$
  - $similarity(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^{n} \omega_k \delta_k s_k(\mathbf{x},\mathbf{y})}{\sum_{k=1}^{n} \omega_k \delta_k}$
  - Can also define a weighted form of distance

$$d(\mathbf{x}, \mathbf{y}) = \left( \sum_{k=1}^{n} w_k |x_k - y_k|^r \right)^{1/r}$$

# Data Exploration

# Data Exploration

**Data exploration** is the preliminary investigation of data to understand its quality, content, and patterns **before** applying any data-mining algorithm.

## Why do it?

- Understand distributions of variables
- Detect unusual values (outliers)
- Discover relationships and patterns
- Assess data quality and ground truth reliability
- Guide preprocessing and modeling choices

### Example: Small Patient Dataset (n = 12)

| ID | Age (years) | Systolic BP (mmHg) | Cholesterol (mg/dL) | Diagnosis (Ground truth) | Lab Test (Observed) |
|---|---|---|---|---|---|
| P01 | 45 | 120 | 200 | No Disease | No Disease |
| P02 | 52 | 130 | 210 | Disease | Disease |
| P03 | 38 | 110 | 180 | No Disease | No Disease |
| P04 | 60 | 140 | 250 | Disease | Disease |
| P05 | — | 135 | 220 | Disease | Disease |
| P06 | 47 | — | 190 | No Disease | No Disease |
| P07 | 71 | 160 | — | Disease | Disease |
| P08 | 34 | 105 | 170 | No Disease | No Disease |
| P09 | 57 | 148 | 260 | Disease | No Disease |
| P10 | 65 | 170 | 310 | Disease | Disease |
| P11 | 29 | 95 | 150 | No Disease | No Disease |
| P12 | 68 | — | 280 | Disease | Disease |

— Missing values

Red text: Disagreement with ground truth

### Missing Values

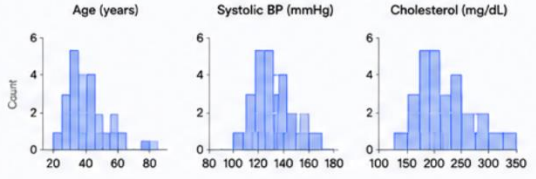
- Age: 1 missing (8.3%)
- Systolic BP: 2 missing (16.7%)
- Cholesterol: 1 missing (8.3%)

How should we handle them?

**Key Question:**

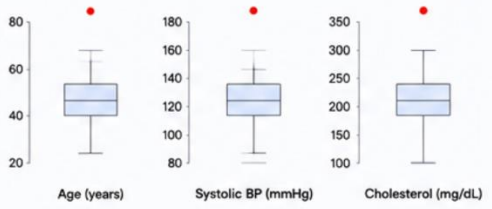
What can we learn about this data **before** building any model?

### 1. Understand Distributions



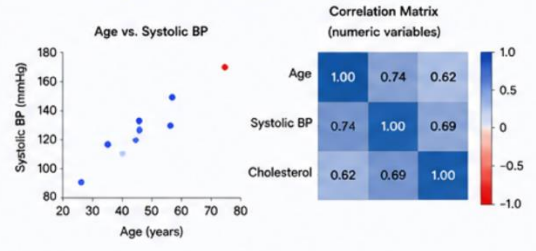

What is the typical range? Are the distributions skewed or multi-modal?

### 2. Detect Unusual Values (Outliers)




Are there extreme values that may be errors or special cases?

### 3. Explore Relationships




Are variables related? Which relationships seem strong?

### 4. Data Quality & Ground Truth (Reliability)

Observed (Lab Test)

| Ground Truth (Ottogenus) | Disease | No Disease | Total |
|---|---|---|---|
| Disease | 5 | 1 | 6 |
| No Disease | 0 | 6 | 6 |
| Total | 5 | 7 | 12 |

**Agreement Metrics**

- Accuracy = 91.7%
- Precision (Disease) = 100%
- Recall (Disease) = 83.3%
- Cohen's kappa = 0.83 (Almost perfect)

How well do the observed labels agree with the ground truth?
Are there labeling errors or inconsistencies?

# Summary Statistics: Understanding the Center

The **mean** (average) uses all values and is affected by extreme values (outliers).
The **median** is the middle value and better represents the "typical" observation when data are skewed or contain outliers.

**Example: Waiting time at clinic (minutes)**

| Patient | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 |
|---|---|---|---|---|---|---|---|---|
| Waiting time | 4 | 5 | 5 | 5 | 6 | 7 | 8 | 45 |

*The value 45 min is an outlier (much larger than the others).*

**Results**

- **Mean (average):** 11.3 min
- **Median:** 6 min
- **Min:** 4 min
- **Max:** 45 min
- **Q1 (25th percentile):** 5 min
- **Q3 (75th percentile):** 7.5 min
- **IQR (Q3 – Q1):** 2.5 min

**Interpretation**

- The mean (11.3) is pulled up by the outlier (45).
- The median (6) better represents the typical waiting time.
- Most patients wait between 5 and 7.5 minutes (IQR).



**When to use what?**

**Use Mean**

- ✓ When the data is roughly symmetric (bell-shaped) and has no extreme outliers.
- ✓ Useful for many statistical methods that assume "average" behavior.

**Use Median**

- ✓ When the data is skewed or contains outliers.
- ✓ More robust – it is not affected much by a few extreme values.



**Key takeaway:** Always look at both mean and median to understand the "center" of your data.
If they are very different, your data may contain outliers or be skewed.

# Summary Statistics: Understanding the Spread

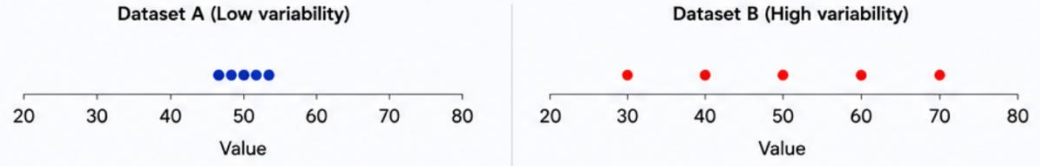
| Dataset A | Dataset B |
|---|---|
| 48, 50, 52, 49, 51 | 30, 50, 70, 50, 60 |
| Mean = 51 \| SD = 1.6 | Mean = 51 \| SD = 14.1 |

**Key insight:**
Both groups have the same average (51), but Dataset B is much more variable.

**Main measures of spread**

- **Range** = max – min
- **Standard Deviation (SD)** – sensitive to outliers
- **Variance** = $SD^2$
- **Interquartile Range (IQR** = Q3 – Q1) – robust to outliers
- **Mean Absolute Deviation (MAD)** – robust to outliers

**Distributions**




- Values are close to the mean (51).
- Values are far from the mean (51) in many cases.

**Robust choice**
Use IQR, MAD or other median-based measures when the data contains outliers or is skewed.

| Dataset | Min | Q1 (25%) | Median | Q3 (75%) | Max | Range (Max – Min) | SD | IQR (Q3 – Q1) | MAD |
|---|---|---|---|---|---|---|---|---|---|
| A (Low variability) | 48 | 49 | 50 | 51 | 52 | 4 | 1.6 | 2 | 1 |
| B (High variability) | 30 | 40 | 50 | 60 | 70 | 40 | 14.1 | 20 | 10 |

**When to use what?**

**Use SD (or Variance):**
- Data is roughly symmetric (bell-shaped) and has no extreme outliers.
- Many statistical models assume this.

**Use IQR or MAD:**
- Data is skewed or contains outliers.
- Need a measure that reflects the typical spread of the majority.

**Use Range:**
- Simple quick check of minimum and maximum.
- But very sensitive to outliers (use with caution).

**Key takeaway:** Spread tells us how much the values vary around the center. Always check spread – two datasets with the same mean can be very different!

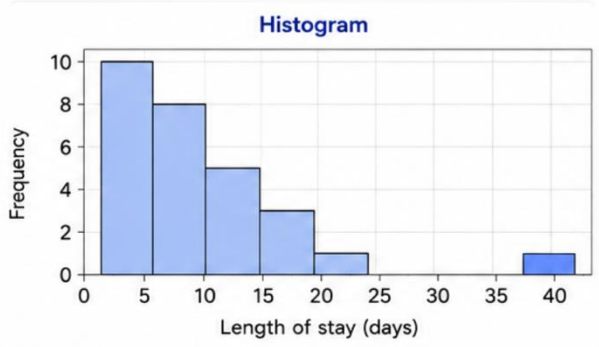
# Understanding Distributions

- Use histograms and box plots to reveal the shape and problems in data

**Example: Length of hospital stay (days)**
(n = 30 patients)

**Data (sorted):**

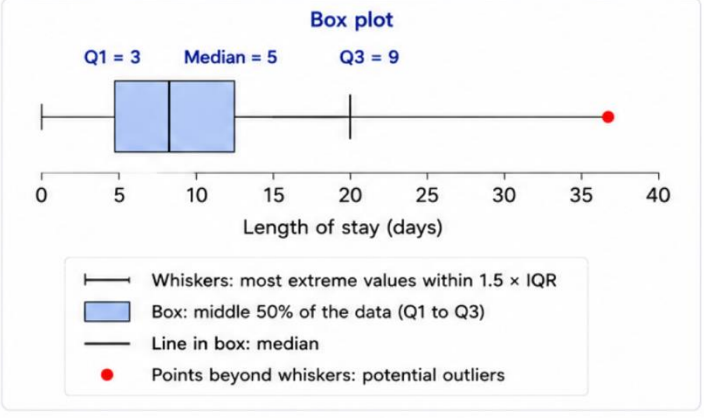
1, 1, 2, 2, 2, 3, 3, 3, 4, 4,
4, 4, 5, 5, 5, 6, 6, 6, 7, 7,
8, 8, 9, 10, 11, 12, 14, 18, 20, 35




**What we learn from the histogram:**

- Most patients stay between 1 and 10 days.
- The distribution is right-skewed (long tail to the right).
- There is a possible outlier around 35 days.

| Summary statistics | |
|---|---|
| Min | 1 |
| Q1 (25%) | 3 |
| Median | 5 |
| Q3 (75%) | 9 |
| Max | 35 |
| IQR (Q3 – Q1) | 6 |
| Mean | 7.8 |
| SD | 7.6 |




**What we learn from the box plot:**

- 50% of patients stay between 3 and 9 days.
- The median stay is 5 days.
- Right whisker is much longer than the left → right-skewed distribution.
- The point at 35 days is an outlier.

**Why it matters?**

Visualizations quickly reveal skewness, unusual values, and data quality issues that may not be obvious from numbers alone.

**Takeaway:** Histograms show the overall shape of the distribution. Box plots summarize the distribution and highlight outliers. Use both to understand your data before applying any data-mining algorithm.

# Exploring Relationships In Data

**Goal:** Understand whether two variables are related, and how strong that relationship is.

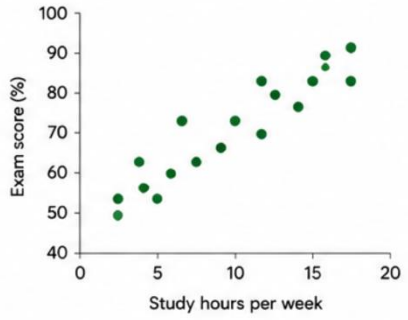
## Example

We collect data on 15 students:

- X = Study hours per week
- Y = Exam score (%)

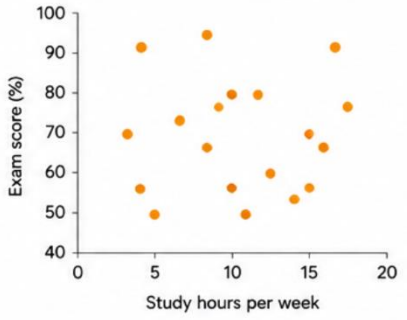
| Statistic | Value |
|---|---|
| Mean (X) | 10.1 |
| Mean (Y) | 74.3 |
| SD (X) | 4.2 |
| SD (Y) | 11.6 |
| Correlation (r) | 0.78 |

### A. Positive correlation (r ≈ +0.78)



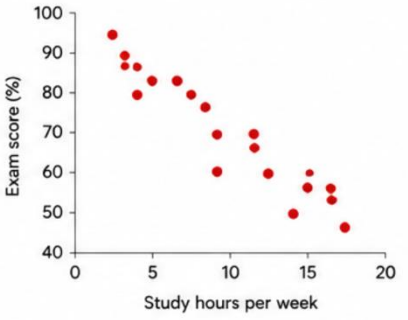

As study hours increase, exam scores tend to increase.
**Strong positive relationship.**

### B. No correlation (r ≈ 0.03)




No clear pattern.
**X and Y do not appear related.**

### C. Negative correlation (r ≈ −0.72)




As study hours increase, exam scores tend to decrease.
**Strong negative relationship.**

## Pearson Correlation Coefficient (r)

Measures the strength and direction of a **linear relationship.**



- $r \in [-1, 1]$
- $r > 0$ : positive relationship
- $r < 0$ : negative relationship
- $r \approx 0$ : no linear relationship
- $|r|$ closer to 1 → stronger relationship

## Interpreting r (rule of thumb)

| \|r\| value | Strength |
|---|---|
| 0.00 – 0.19 | Very weak |
| 0.20 – 0.39 | Weak |
| 0.40 – 0.59 | Moderate |
| 0.60 – 0.79 | Strong |
| 0.80 – 1.00 | Very strong |

## Points to remember

- Correlation measures linear relationship only. It does NOT imply **causation.**
- Outliers can strongly affect *r*. Always check the scatter plot!
- Different patterns (curved, clusters) may exist even when $r \approx 0$.

**Key takeaway:** Scatter plots help visualize relationships and detect patterns or outliers. Correlation quantifies linear relationships, but it is not the whole story.

# Data Preprocessing

# Data Preprocessing

- Data Integration & Aggregation
- Sampling
- Data Cleaning
- Attribute Transformation
- Normalization, standardization
- Feature selection
- Feature creation

# Data Integration and Aggregation

- Integration combines data from multiple sources to create a unified dataset
- Aggregation summarizes multiple attributes or objects into a single, higher-level attribute or object

## Aggregation

Combining two or more attributes (or objects) into a single attribute (or object).

### Purpose

**Data reduction**
Reduce the number of attributes or objects.

**Change of scale**
- Cities aggregated into regions, states, countries, etc.
- Days aggregated into weeks, months, or years.

**More "stable" data**
Aggregated data tends to have less variability.

## Examples

### 1. Aggregating Cities into Regions

| City | Sales (KSEK) |
|---|---|
| Växjö | 120 |
| Lund | 90 |
| Malmö | 110 |
| Göteborg | 150 |
| Umeå | 80 |

→

| Region | Total Sales (KSEK) |
|---|---|
| South | 320 |
| West | 150 |
| North | 80 |

Cities are aggregated to their regions by summing sales.

### 2. Aggregating Days into Months

| Date | Temperature (°C) |
|---|---|
| 2025-04-01 | 12.1 |
| 2025-04-02 | 13.0 |
| ... | ... |
| 2025-04-30 | 15.3 |

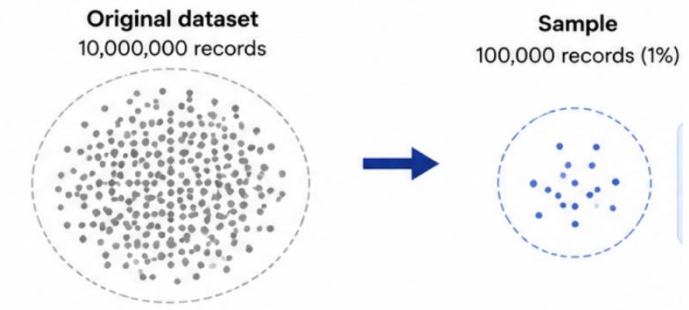
→

| Month | Avg. Temperature (°C) |
|---|---|
| Apr 2025 | 13.2 |
| May 2025 | 16.7 |
| Jun 2025 | 19.4 |

Daily temperatures are aggregated into monthly averages.

**Key idea:** Aggregation reduces complexity and reveals higher-level patterns, but may hide important details if the level of aggregation is too coarse.

# Sampling




- **Sampling** selects a subset of data that represents the larger dataset.
  - Reduce computational cost for very large datasets
  - Explore data quickly before running full analysis
  - Develop and test models faster
  - Data collection itself may be expensive or time-consuming
- Using a sample will work almost as well as using the entire data set, if the sample is **representative**
- A sample is **representative** if it has approximately the same properties (of interest) as the original set of data

# Types of Sampling and Sampling Size

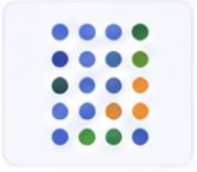
## Types of Sampling

Different sampling methods are used depending on the data and the goal.



### 1. Simple Random Sampling

- Every record has an equal chance of being selected.
- Easy to implement; requires a complete list of the data.



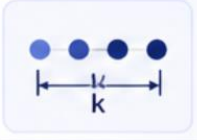
### 2. Stratified Sampling

- The dataset is divided into groups (strata) based on a characteristic (e.g., class, region, age).
- A random sample is taken from each stratum.
- Ensures that all important groups are represented.



### 3. Cluster Sampling

- The data is divided into clusters (e.g., schools, cities).
- Randomly select some clusters and include all records within those clusters.
- Cost-effective when data is naturally grouped.




### 4. Systematic Sampling

- Select every $k$-th record after a random starting point.
- Simple and fast when the data has no hidden pattern.



### 5. Convenience / Judgment Sampling

- Select the most accessible records or those judged most useful.
- Quick but may introduce bias (non-representative).

## How to Choose a Sample Size?

Larger samples give **more** accurate and reliable results, but require more time and resources.

**Common factors that affect sample size:**

- Size and variability of the population
- Desired confidence level (e.g., 95% or 99%)
- Acceptable margin of error (precision)
- Type of analysis or model to be built
- Available time and resources

**Rule of thumb (for estimating proportions)**

For a large population, the required sample size $n$ can be approximated as:

$$n = \frac{Z^2\, p\,(1-p)}{E^2}$$

where:

Z = Z-score for the chosen confidence level (1.96 for 95%)

p = estimated proportion (use 0.5 if unknown → gives maximum $n$)

E = desired margin of error (e.g., 0.05 for ±5%)

**Remember:** The best sample is not only large enough, but also representative of the population.

# Data Cleaning

## Purpose in this process

**We do not need to fix the data now.**

1. **Identify and count problematic records**
   Understand how many issues exist and of what type.
2. **Decide what to do with them**
   Choose the most appropriate action(s) based on the problem and the goal of the analysis.

**What we decide**
For each type of problem, decide what action to take before analysis or modeling.

## Techniques to Detect Data Problems

| Type of Problem | How to Detect (Examples) |
|---|---|
| Missing values | • Count missing (per attribute, per record)<br>• Visualize missing patterns |
| Invalid data / Constraint violations | • Check data types, ranges, formats<br>• Check domain rules (e.g., age $\geq 0$) |
| Outliers / Unusual values | • Use summary statistics (min, max, IQR, z-score)<br>• Box plots, scatter plots |
| Inconsistencies (within data) | • Check logical rules (e.g., start date $\leq$ end date)<br>• Compare related fields |
| Duplicates | • Exact duplicates<br>• Near-duplicates (e.g., same person with minor differences) |
| Incomplete records | • Records with too many missing or critical fields empty |

**Output of this step:** A summary report with counts of problematic records by type.

## Options: What Can We Do with Problematic Records?

| Option | When to Use | Pros | Cons |
|---|---|---|---|
| Remove (Delete) | Few records, severe errors, or not important for analysis | • Simple<br>• Avoids misleading results | • May lose information or reduce sample size |
| Correct (Fix) | Errors are clear and can be reliably fixed | • Keeps data<br>• Improves quality | • Risk of introducing bias<br>• May be time-consuming |
| Impute (Fill in) | Missing values, when information can be estimated | • Keeps records<br>• Preserves sample size | • Introduces uncertainty<br>• Choice of method matters |
| Mark / Flag | Uncertain cases that should be tracked | • Preserves all data<br>• Keeps information about quality | • Requires downstream handling<br>• May complicate analysis |
| Keep as is | Issue has minor impact on the analysis | • No extra work<br>• Keeps all data | • Problem may influence results |

# Attribute Transformation

## Why Transform Attributes?

**Improve algorithm performance**
Many algorithms assume specific data types or scales.

**Reduce complexity**
Simpler representations are easier to understand and model.

**Capture useful patterns**
Reveal relationships that are hidden in the original form.

**Meet assumptions**
Satisfy requirements such as numeric input, linearity, etc.

## General Principles

- Preserve as much useful information as possible.
- Make data easier to interpret and model.
- Avoid introducing unnecessary distortion.
- Keep the transformation simple and explainable.

## Common Attribute Transformation Techniques

### 1. Binarization

Convert an attribute into two values (0/1, False/True).

**Example: Has Disease**

Before

| Patient | Disease |
|---|---|
| P1 | Yes |
| P2 | No |
| P3 | Yes |
| P4 | No |

→

After

| Patient | Disease (0/1) |
|---|---|
| P1 | 1 |
| P2 | 0 |
| P3 | 1 |
| P4 | 0 |

**Use:** Convert categorical or condition-based attributes.

### 2. Discretization (Binning)

Convert a continuous attribute into a set of intervals or bins.

**Example: Age**

Before

| Age (years) |
|---|
| 23 |
| 31 |
| 45 |
| 62 |
| 70 |
| 85 |

→

After

| Age Group (bins) |
|---|
| 18–30 |
| 31–40 |
| 41–50 |
| 51–60 |
| 61–70 |
| 71+ |

**Use:** Handle skewed distributions, reduce sensitivity to noise, simplify.

### 3. Text to Numbers

Convert text/categorical values to numeric form.

**Example: City**

Before

| City |
|---|
| Stockholm |
| Göteborg |
| Malmö |

→

After

| Label Encoding (integer) |
|---|
| 2 |
| 0 |
| 1 |

or

| One-Hot Encoding | | |
|---|---|---|
| Stockholm | Göteborg | Malmö |
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

**Use:** Required for algorithms that need numeric input.

### 4. Other Representations

Represent an attribute in a different form to capture meaning better.

**Examples:**

- **Ordinal encoding:** Low < Medium < High → 1, 2, 3
- **Frequency/Count encoding:** Category → frequency
- **Target/Mean encoding:** Category → mean(target)
- **Date/Time transformations:**
  - Extract components (year, month, day, hour)
  - Cyclical encoding (e.g., sin/cos for hour of day)
- **Log / Power transformation:** Reduce skewness
- **Vector representations:** Word embeddings, TF-IDF vectors, etc.

**Use:** Capture relationships, reduce dimensionality, or better model complex patterns.

## Guidelines

- Choose transformations based on data type and analysis goal.
- Apply simple transformations first.
- Validate that the transformation improves model performance and interpretability.

## Before vs After – Example

**Original Data**

| ID | Age | Income | City | Has Card | Join Date |
|---|---|---|---|---|---|
| 1 | 23 | 12500 | Stockholm | Yes | 2023-01-15 |
| 2 | 45 | 72000 | Göteborg | No | 2023-06-20 |
| 3 | 62 | 54000 | Malmö | Yes | 2023-03-10 |

→

**Transformed Data (example)**

| ID | Age Group | Income (log) | City (one-hot) | | | Has Card (0/1) | | Month | Day of Week |
|---|---|---|---|---|---|---|---|---|---|
| | | | Sthlm | Gbg | Malmö | | | | |
| 1 | 18–30 | 9.43 | 1 | 0 | 0 | 1 | 1 | 15 | Sun |
| 2 | 41–50 | 11.19 | 0 | 1 | 0 | 0 | 1 | 6 | Tue |
| 3 | 61–70 | 10.90 | 0 | 0 | 1 | 3 | 3 | 10 | Fri |

# Normalization, Standardization

Normalization and standardization are techniques to put numeric attributes on a **comparable scale**.
They are important when using distance measures, similarity measures **or algorithms** that are sensitive to the scale of the data.

## When to use which?

### Normalization (min–max scaling)

**Use when:**

- You need all features to be in the same range (e.g., 0–1).
- The data does not have extreme outliers.
- Useful for distance-based methods (e.g., k-NN, clustering) and neural networks.

### Standardization (z-score scaling)

**Use when:**

- You want features to have mean 0 and standard deviation 1.
- The data may contain outliers.
- Useful for algorithms assuming normality (e.g., linear regression, PCA) and many ML models.

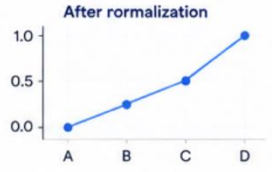
## How it works

### Normalization (min–max scaling)

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$x'$ – normalized value
$x$ – original value
$x_{max}$, $x_{min}$ – max and min of the feature

| Value | Original | Normalized (0–1) |
|---|---|---|
| A | 10 | 0.00 |
| B | 20 | 0.25 |
| C | 40 | 0.75 |
| D | 50 | 1.00 |

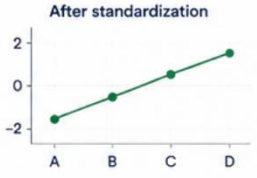



### Standardization (z-score)

$$z = \frac{x - \mu}{\sigma}$$

$z$ – standardized value
$\mu$ – mean
$\sigma$ – standard deviation

| Value | Original | Standardized (z-score) |
|---|---|---|
| A | 10 | -1.16 |
| B | 20 | -0.39 |
| C | 40 | 0.77 |
| D | 50 | 1.55 |

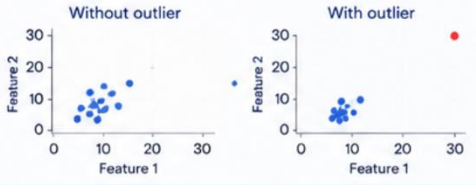



## Robustness to outliers

### Normalization

**Less robust**
Outliers can strongly affect the min and max, and thus the scaling.

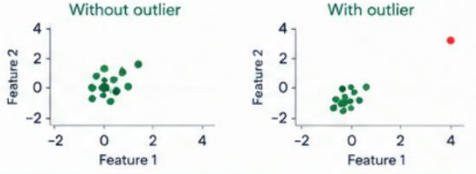
**Example: Effect of an outlier**




### Standardization

**More robust**
Uses mean and standard deviation, which are less sensitive to extreme values.

**Example: Effect of an outlier**




## Key takeaways

- Use **normalization** when you need a fixed range (e.g., 0–1) and the data has no extreme outliers.
- Use standardization when you want mean 0 and unit variance, and the data may contain outliers.
- Both methods improve the performance of many algorithms, especially those based on distances or similarities.

# Feature Selection

Another way to reduce the dimensionality of data:
Select a **subset** of the original features that **keeps the important information** and removes what is not needed.

## Why select a subset of features?

**Reduce complexity**
Fewer features → simpler models, faster training and less storage.

**Improve performance**
Removing useless or redundant features can increase accuracy and generalization.

**Better interpretability**
Models are easier to understand and explain with fewer, relevant features.

**Avoid problems**
Reduce noise and the risk of overfitting, especially when the number of features is large.

## Types of Features to Remove

**Redundant features**
- Duplicate much or all of the information contained in one or more other attributes.
- **Example:** purchase price of a product and the amount of sales tax paid.

**Irrelevant features**
- Contain no information that is useful for the data mining task at hand.
- **Example:** students' ID is often irrelevant to the task of predicting students' GPA.

**Goal:** Keep the features that matter, remove the ones that don't.

## How can we select a subset?

**Filter Methods** (independent of the model)
- Use statistical measures to score features.
- Select top-k features.
- Examples: correlation, chi-square, mutual information, variance threshold.

OR

**Wrapper / Embedded Methods** (use the model)
- Evaluate feature subsets using a model's performance.
- Search for the best subset.
- Examples: forward selection, backward elimination, LASSO, tree-based importance.

**Remember:** Dimensionality reduction (e.g., PCA) is another approach that transforms features into fewer new ones.
Feature subset selection keeps a subset of the original features.



# Feature Creation

**Feature creation** is the process of creating **new attributes (features)** from existing data to capture important information more effectively than the original attributes.

## Why create features?

**Improve predictive power**
New features can reveal patterns that original features cannot.

**Simplify models**
Good features make models simpler, faster and more interpretable.

**Better generalization**
Well-designed features reduce noise and help models generalize to new data.

**Domain knowledge leverage**
Using what we know about the problem helps create meaningful features.

## When is it especially important?

**Especially useful when:**

- You have a **small dataset**
- Raw data is high-dimensional or noisy
- Important information is hidden in complex relationships
- You want to improve model performance and interpretability

### Examples

**Date & Time**

| Original | New features |
|---|---|
| 2024-05-01 13:45 | Day of week = Wed |
| 2024-05-02 09:15 | Hour = 9 |
| 2024-05-03 18:30 | Is weekend = No |
| ... | ... |

**Transaction Data**

| Original | New features |
|---|---|
| Price = 120 | Price per unit |
| Quantity = 3 | Total amount = 360 |
| Discount = 10% | Price after discount |
| ... | ... |

**Geographic Data**

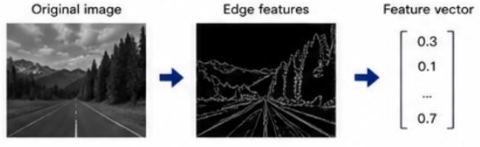
| Original | New features |
|---|---|
| Latitude, Longitude | Distance to center |
| ZIP code | Region |
| Address | Urban / Rural |
| ... | ... |

## Three General Methodologies

### 1 Feature Extraction

Transform raw data into a compact representation that captures the most important information.

**Example:** *extracting edges from images*




**Common in:** images, audio, text, signal processing.

### 2 Feature Construction

Combine or transform existing attributes using mathematical or logical operations.

**Example:** *dividing mass by volume to get density*

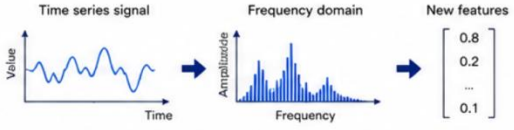
| Original features | | Constructed feature |
|---|---|---|
| Mass (kg) | Volume ($m^3$) | Density ($kg/m^3$) = Mass / Volume |
| 10 | 2 | 5.0 |
| 15 | 3 | 5.0 |
| ... | ... | ... |

**Common in:** ratios, differences, interactions, aggregations, polynomial features.

### 3 Mapping Data to New Space

Map data to a new representation (space) where patterns are more easily captured.

**Example:** *Fourier and wavelet analysis*




**Common in:** signal processing, text (embeddings), computer vision, dimensionality mapping.

### Key takeaways

- Good features are the key to good models.
- Use domain knowledge + data insights.
- Create features, evaluate them, and keep what helps.

### Tip

Not all created features are useful. Always validate their impact with cross-validation or hold-out data.

# Summary



## Understand your data

- Know your objects (records) and attributes (variables).
- Examine data types: numeric, categorical, temporal, etc.
- Get familiar with the dataset structure and basic content.



## Check your data

- Identify missing values and decide how to handle them.
- Detect errors, duplicates, noise and outliers.
- Assess data quality and how reliable the ground truth is.



## Compare and explore

- Use summary statistics to understand center and spread.
- Use visualizations (histogram, box plot, scatter plot, etc.) to see distributions, patterns and relationships.
- Look for unusual values, trends and correlations.



## Prepare your data

- Clean: handle missing values, outliers and inconsistencies.
- Transform: normalize/scale, encode, aggregate, etc.
- Select and create features that are informative.
- Good preparation leads to better models and insights.